

LOO WEE LUN

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EDUCATION

Nanyang Technological University, Singapore

Aug 2022 – Jun 2026

Bachelor of Science in Mathematical Sciences, Honours (Highest Distinction), Specialisation in Pure Mathematics

- CGPA: 4.86/5.00
- Relevant Coursework: Measure Theory, Complex Analysis, Algebraic Topology, Module Theory & Homological Algebra, Graph Theory, Combinatorics, Cryptography, Differential Equations.
- Completed two modules outside the standard curriculum: MH7002 (Discrete Methods, a graduate module on network flow theory) and MH4930 (Special Topic on Introduction to Homological Algebra).
- Award: ASEAN Undergraduate Scholarship

ACADEMIC PROJECTS

MH4900: Final Year Project

Aug 2025 – Apr 2026

Topic: Mathematical Analysis of Trading Strategies

Supervisor: Prof. Bernhard Schmidt

- Studied technical analysis fundamentals and implemented a backtesting framework in code to evaluate classical trading strategies.
- Proposed a novel technical indicator with an accompanying computation algorithm; the indicator reproduces ZigZag-like output while being substantially simpler to compute and free of the repainting issue common to existing indicators. Grade: A.

MH4921: Supervised Independent Study II

Jan 2026 – Apr 2026

Topic: Further Investigation in Homological Algebra

Supervisor: Dr. Lim Kay Jin

- Studied homological algebra on abelian categories, including foundational category theory, group cohomology, the Ext and Tor functors, and spectral sequences, following Weibel.
- Authored a final report examining triangulated categories and derived categories; awarded an A+.

Odyssey Research Programme

May 2025 – Jul 2025

Topic: Fourier Transform on LCA Groups

Supervisor: Dr. Tang Wee Kee

- Studied abstract Fourier analysis on locally compact abelian (LCA) groups following Rudin.
- Developed and presented a research poster at a symposium, communicating the theory to a broader audience.

PS5916: Undergraduate Research Experience on Campus (URECA)

Aug 2024 – Jun 2025

Topic: A New Class of Cyclic Codes

Supervisor: Prof. Bernhard Schmidt

- Proposed a new class of codes inspired by quadratic residue codes and used Magma to compute minimum weight bounds for the class.
- Conjectured that the proposed codes satisfy the square-root bound; findings documented in a written report.

MH4910: Summer Research Experience

May 2024 – Jul 2024

Topic: Investigation on Schubert Polynomials

Supervisor: Dr. Lim Kay Jin, co-supervised by Dr. Ku Cheng Yeaw

- Studied foundational algebraic combinatorics, including symmetric groups, Bruhat order, and combinatorial diagrams, to construct Schubert polynomials.
- Computed the Schubert polynomial for S_5 using SageMath and documented findings in a written report; awarded an A+.

PS5915: Undergraduate Research Experience on Campus (URECA)

Aug 2023 – Jun 2024

Topic: T_2 -norms of Generators of Principal Ideals in Cyclotomic Fields

Supervisor: Prof. Bernhard Schmidt

- Investigated principal ideals in cyclotomic fields, proved that a generating ideal forms a cyclic code, and applied a minimum-weight bound from coding theory to bound its T_2 -norm.
- Proposed a combinatorial question to further strengthen the result; presented findings at the International Conference of Undergraduate Research (ICUR) 2024 and authored a written report.

MH4920: Supervised Independent Study I

Jan 2024 – Apr 2024

Topic: Algebraic Graph Theory

Supervisor: Dr. Gary Royden Watson Greaves

- Studied spectral graph theory following Godsil and Royle, emphasizing matrix theory, strongly regular graphs, equiangular lines, and two-graphs.
- Authored a report including a short survey of open questions in the area; awarded an A+.

Odyssey Research Programme

May 2023 – Jul 2023

Topic: Introduction to Analytic Number Theory

Supervisor: Dr. Gary Royden Watson Greaves

- Built up prerequisite theory and studied the elementary proof of the Prime Number Theorem.
- Developed and presented a research poster at a symposium.

WORK EXPERIENCE

Undergraduate Teaching Assistant

Aug 2024 – Apr 2026

School of Physical and Mathematical Sciences, Nanyang Technological University

- Led tutorial sessions for 7 undergraduate classes in Linear Algebra I, Discrete Mathematics, and Calculus III, primarily for Mathematical Sciences students.
- Identified and addressed recurring student misconceptions in each topic through targeted in-session explanations.
- Prepared and distributed structured post-tutorial summaries covering key concepts, worked solutions, and exam-relevant insights.
- Achieved a 93.9% weighted average teaching rating across 71 student feedback responses.

Peer Tutor

Aug 2024 – Nov 2025

School of Physical and Mathematical Sciences, Nanyang Technological University

- Tutored 20+ first- and second-year Mathematical Sciences undergraduates in individual and small-group sessions.
- Used a problem-first approach, working through problems collaboratively rather than presenting solutions directly.
- Provided targeted exam preparation and study planning ahead of mid-terms and finals.
- Mentored students retaking previously failed courses through to successful completion.

Tuition Tutor

May 2025 – Jul 2025

Learning Smart

- Instructed 10 Primary 1 and 2 students in small-group settings of up to three students.
- Designed lesson activities suited to Primary 1–2 attention spans to sustain engagement across sessions.
- Tracked individual progress and adjusted teaching pace and methods accordingly.

Daycare Teacher

Feb 2022 – Jun 2022

Harmonious Music

- Supervised after-school care and academic support for 20+ Primary 1 students, including routines, homework guidance, and tuition sessions.
- Developed original instructional materials for homework support and tuition sessions.

CO-CURRICULAR ACTIVITIES

NTU Hall 14 Contract Bridge Club - Captain

Aug 2024 – Jul 2025

(Member: Aug 2023 – Jul 2024)

- Hosted weekly training sessions as club captain, teaching strategic techniques to new members and enhancing team skills.
- Directed team strategy and coordination, achieving a Top 8 placement in an inter-school competition.
- Contributed as a member to the club's championship win in a prior inter-school tournament.

Mathematics Subject Representative – Subcommittee

Aug 2022 – Jul 2024

NTU Students' Physical and Mathematical Sciences Club

- Designed, organized, and hosted annual Estimathon events, facilitating engagement and problem-solving activities.
- Developed original questions and structured sessions to ensure smooth execution, successfully delivering the program twice over two years.

SKILLS

Languages: Chinese (Native), English (Proficient), Malay (Proficient)

Digital Skills: Microsoft Office Suite, LaTeX, Python, R, Magma, PariGP